

ADDIS ABABA UNIVERSITY
COLLEGE OF HEALTH SCIENCES
SCHOOL OF PHARMACY



Bachelor of Pharmacy
(BPharm) Curriculum

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ADDIS ABABA
ETHIOPIA

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1. Background

In the past few decades, Ethiopia went through numerous changes that influenced the different sectors in the economy. These changes have positively impacted the way the health system operates and improved major indicators of health care in the country. However, the country still remains to be characterized by its low socioeconomic status and health service coverage. As a result, there is low ratio and improper mix of health professional to the population. The Ethiopian health policy is based on health promotion and disease prevention with focus on ensuring equity. The policy gives priority to ensuring access for the rural and unprivileged urban population. Successful implementation of this policy is highly dependent on availability of well-trained health professionals, both in quality and quantity. In this regard, the pharmacy service as an essential component of the health care delivery system requires properly trained professionals.

While pharmacists' vital role remains to be dispensing of medications and devices and ensuring appropriate therapy and outcomes, they also work in the areas of health promotion and disease prevention, pharmaceutical production and health systems management. In this regard, pharmacists provide their services in a variety of settings in response to a dynamic and evolving set of primarily local health care priorities and needs. There are also regional, national and international policies and factors, which dictate the need for developments in pharmacy practice. Within this context, pharmacists are medication experts in the treatment of disease and in health promotion. This expertise, in its broadest sense, encompasses the preparation, supply and control of medicinal products and assurance of desired outcomes of treatment by medication. It thus begins with the medicine development process and continues through to medication's ultimate benefit to the individual and to society generally. This expertise has its foundations in the pharmaceutical sciences and related research, and has its focus on the individual and populations.

The needs of modern health care systems require that the role of the pharmacist develop rapidly to meet its demands. It is also expected that pharmacists' roles may evolve with time, changes in societal needs and technological progress. Thus, the pharmacy curriculum should change with such developments in order to address the needs of the society. Generally, the pre-service education is designed to ensure that the newly qualified pharmacist has the necessary knowledge and skills to commence practicing competently in a variety of settings including community and hospital pharmacy and the pharmaceutical industry. As such, the curriculum used in pre-service education needs to be responsive to changes in the practice setting, the needs of the society and knowledge

and technological innovations. That is why a robust evaluation of the current performance of the curriculum and a constant formative evaluation of progresses in the field of pharmacy are highly recommended. In line with such recommendations, this curriculum document was prepared after a comprehensive assessment of the performance of the previous curriculum and needs for revision.

2. Rationale

The global landscape of pharmacy practice and thus the education of pharmacists have evolved significantly in the past three decades. The main driver of such changes has been the advent of patient-centered pharmaceutical care as a philosophy of pharmacy practice. As such, universities in Ethiopia have been trying to increasingly incorporate more clinical courses in the existing pharmacy curriculum throughout the past decade. First, Ethiopian universities started offering a clinical oriented curriculum with significantly higher number of clinical-leaning courses in 2008, which started graduating students in 2013. In this endeavour the universities have tried very hard to maintain a balance between the clinical and non-clinical course contents in a way that reflects the Ethiopian context. The same year, the graduates were deployed to the different practice setting in Ethiopia, there was a move to modularize the curriculum in a nationally harmonized manner. This was done by clustering courses into general umbrellas of modules, without the significant modification of course contents in the previous curriculum.

It is generally recommended that higher education curriculums should undergo major revisions at least every five years. As it is a little more than five years since the implementation of the current curriculum, it was imperative to initiate its revision. In addition, the then Ministry of Science and Higher Education (MOSHE), which has since been remerged with the Ministry of Education (MOE), has prescribed certain common courses to be included in all undergraduate curriculums as per the national Educational roadmap spanning from 2020 to 2030. MOSHE also changed the direct placement of candidates, who pass the preparatory school leaving examination, to pharmacy schools. Now students have to take freshman courses in the first semester and join pharmacy schools in the second semester of the first year. Moreover, a nationwide assessment of the performance of the current curriculum including all stakeholders from current students and instructors to graduates and employees indicated the need for revision of the curriculum. Thus, the preparation of this curriculum document was initiated in a bid to respond to these demands for revision.

3. Mission

Our mission is to train highly qualified pharmacists who meet global standards and local healthcare needs focusing on patient-centered care, equipping graduates with the knowledge and skills to excel in diverse pharmaceutical settings, including hospitals, community pharmacies, industry, public health, and research.

4. Objectives

4.1. General Objectives

- The general objective of the B. Pharm Program is to train highly qualified pharmacists who fulfil the essential, minimum common expectations of health care systems worldwide while fulfilling local needs.
- Graduates would be expected to have the required knowledge, skill, attitude and behaviour to support the health care system and to respond to patient needs working in hospital and community pharmacies; drug manufacturing, import and wholesale companies; public health services; academic institutions; pharmaceutical research; drug regulatory body and quality control organizations.

4.2. Specific Objectives

- To train manpower that is more patient-oriented while still having a broad pharmaceutical knowledge to be able to easily adapt to working in any of the settings in the country's pharmaceutical sector.
- To provide practice-based training so that future pharmacists acquire problem-solving skills.
- To facilitate and assist in the transfer and adaptation of pharmaceutical knowledge.

Demand for the program

In Ethiopia, health institutions (hospitals, health centers, NGOs, and others), higher institutions (government and private), research institutions and pharmaceutical industries that require competent pharmacy professionals at different level of training and expertise are progressively growing. As a result, ample opportunities are created for pharmacy practitioners to play a vital role in the country's development. However, the input of these professionals to a great extent depends on the quality of education (training) they receive in the higher learning institutions or universities.

The curriculum review has taken into consideration the existing socio-economic demands, government policies and global experience in training pharmacists. During the revision of the curriculum feedback from stakeholders, former graduates of different universities, and internal and

external assessors were given serious consideration. The global experience in pharmacy curriculum design and implementation was also considered.

Professional Profile

To be effective health care team members, pharmacists need skills and attitudes enabling them to assume many different functions. The concept of the “seven-star pharmacist” was introduced by WHO and taken up by FIP in 2000 in its policy statement on Good Pharmacy Education Practice to cover these roles: caregiver, decision-maker, communicator, manager, life-long learner, teacher and leader. The function of the pharmacist as a researcher was later on added.

These roles of the pharmacist are described below and include the following functions:

Caregiver: Pharmacists provide caring services. They must view their practice as integrated and continuous with those of the health care system and other health professionals. Services must be of the highest quality.

Decision-maker: The appropriate, efficacious, safe and cost-effective use of resources (e.g., personnel, medicines, chemicals, equipment, procedures, and practices) should be the foundation of the pharmacist’s work. At the local and national levels, pharmacists play a role in setting medicines policy. Achieving this goal requires the ability to evaluate, synthesize data and information and decide upon the most appropriate course of action.

Communicator: The pharmacist is in an ideal position to provide a link between prescriber and patient, and to communicate information on health and medicines to the public. He or she must be knowledgeable and confident while interacting with other health professionals and the public. Communication involves verbal, non-verbal, listening and writing skills.

Manager: Pharmacists must be able to manage resources (human, physical and financial) and information effectively; they must also be comfortable being managed by others, whether by an employer or the manager/leader of a health care team. More and more, information and its related technology will provide challenges as pharmacists assume greater responsibility for sharing information about medicines and related products and ensuring their quality.

Life-long-learner: It is impossible to acquire in pharmacy school all the knowledge and experience needed to pursue a life-long career as a pharmacist. The concepts, principles and commitment to life-long learning must begin

while attending pharmacy school and

must be supported throughout the pharmacist's career. Pharmacists should learn how to keep their knowledge and skills up to date.

Teacher: The pharmacist has a responsibility to assist with the education and training of future generations of pharmacists and the public. Participating as a teacher not only imparts knowledge to others, it offers an opportunity for the practitioner to gain new knowledge and to fine-tune existing skills.

Leader: In multidisciplinary (e.g., team) caring situations or in areas where other health care providers are in short supply or non-existent the pharmacist is obligated to assume a leadership position in the overall welfare of the patient and the community. Leadership involves compassion and empathy as well as vision and the ability to make decisions, communicate, and manage effectively. A pharmacist whose leadership role is to be recognized must have vision and the ability to lead.

Researcher: The pharmacist must be able to use the evidence base (e.g., scientific, pharmacy practice, health system) effectively in order to advice on the rational use of medicines in the health care team. By sharing and documenting experiences, the pharmacist can also contribute to the evidence base with the goal of optimizing patient care and outcomes. As a researcher, the pharmacist is able to increase the accessibility of unbiased health and medicines-related information to the public and other health care professionals

4. Graduate Profile

It is envisaged that pharmacy graduates with the B.Pharm. Degree will be capable to assume the following responsibilities and attributes:

5. Core Graduate Competencies

Organize and control the manufacturing, compounding and packaging of pharmaceutical products;
Organize the selection, procurement, storage, and distribution of pharmaceutical materials and products;
Provide Pharmaceutical Care and Dispense and ensure the optimal use of medicines by the patient;
Provide pharmacist-initiated care to patients and ensure the optimal use of medicines;
Provide education and information on health care and medicines;

Promote community health and provide related information and advice; and

Conduct in research to ensure the optimal use of medicines.

Demonstrate a high level of professional ethics in order to satisfy the pharmaceutical needs of the society.

Maintain and expand knowledge through self-directed learning.

Be able to work as a member of the health team.

Possess the necessary background to pursue further advanced study in the pharmaceutical sciences.

The core competencies have the following knowledge, attitude and skill attitude components in the various domains of pharmacy practice.

Domains	Competencies
Patient care	Knowledge:
	• Possess the necessary background to pursue further advanced study in the pharmaceutical sciences • Provide pharmacist-initiated care to patients and ensure the optimal use of medicines • Provide education and information on health care and medicines
	Attitude:
	• Advocate the proper use of necessary materials by screening • Recognize, adhere to and promote established safety rules • Respectful and compassionate to patients, their relatives and other professionals • Be able to work as a member of the health team
	Skill:
	• Provide Pharmaceutical Care and Dispense and ensure the optimal use of medicines by the patient • Communicate effectively both verbally and in writing • Collect, document, retrieve and interpret data related to all their activities clearly and safely
	Knowledge:

Pharmaceutical technology	• Possess the necessary background to pursue further advanced study in the pharmaceutical sciences
	Attitude:
	• Advocate the proper use of necessary materials by screening • Recognize, adhere to and promote established safety rules • Institute and promote safety, quality control and quality assurance in their allowed work area
	Skill:
	• Organize and control the manufacturing, compounding and packaging of pharmaceutical products • Monitor and maintain proper functioning of necessary equipment/reagents • Develop and modify laboratory procedures
Health promotion	Knowledge:
	• Provide education and information on health care and medicines • Promote community health and provide related information and advice
	Attitude:
	• Advocate the proper use of necessary materials by screening
	Skill:
	• Prepare educational materials to promote the rational use of medicines and medical devices • Screen for drug abuse and refer patients to appropriate care centre
Professionalism	Knowledge:
	• Demonstrate a high level of professional ethics in order to satisfy the pharmaceutical needs of the society • Maintain and expand knowledge through self-directed learning
	Attitude:
	• Maintain the pharmacy ethical code of conduct standards • Contribute to stewardship of their profession

	• Recognize, adhere to and promote established safety rules • Pursue graduate training in pharmacy and other health related disciplines • Participate in policy, professional standards, continuing professional development issues pertaining to pharmacy profession • Respectful and compassionate to patients, their relatives and other professionals
	Skill:
	• Communicate effectively both verbally and in writing • Engage in policy, professional standards, and continuing professional development issues pertaining to pharmacy profession
Pharmacy law and regulatory affairs	Knowledge:
	• Familiarity with the latest laws, directives and guidelines governing the pharmaceutical sector
	Attitude:
	• Maintain the pharmacy ethical code of conduct standards • Recognize, adhere to and promote established safety rules
	Skill:
	• Store and use laboratory supplies and dispose expired drugs safely according to the rules and regulations • Develop and modify laboratory procedures
Leadership and management	Knowledge:
	• Use his/her critical thinking to improve the pharmacy working environment • Familiarize him/her with latest scientific findings to improve the quality of services rendered to the society
	Attitude:
	• Institute and promote safety, quality control and quality assurance in their allowed work area

	• Participate in policy, professional standards, continuing professional development issues pertaining to pharmacy profession
	Skill:
	• Organize the selection, procurement, storage, and distribution of pharmaceutical materials and products • Plan drugs and equipment logistic procurement, evaluation, setup and auditing • Monitor the inventory in storage, work and laboratory area • Provide professional services, leadership and quality assurance in work areas • Demonstrate leadership and management skills
Scholar (Research and Evidence Based Practice)	Knowledge:
	• Familiarize him/her with latest scientific findings to improve the quality of services rendered to the society • Maintain and expand knowledge through self-directed learning • Participate in research to ensure the optimal use of medicines
	Attitude:
	• Pursue graduate training in pharmacy and other health related disciplines
	Skill:
	• Participate and/or conduct research and development of new drugs discovery technologies • Develop and modify laboratory procedures

6. Admission Requirements

6.1. Regular Program

Admission requirements will be as per the University Senate Legislation. The University shall admit for undergraduate studies:

- Students who have completed Ethiopian high school education and satisfy its entrance qualification assessments.
- Applicants with foreign high school education of equivalent level to the Ethiopian high school education, as determined by the Ministry of Science and Higher Education, and who satisfy entrance requirements
- Considering the extensive practical attachments of the program, students with major disabilities (vision, mobility and hearing-related problems or those who cannot effectively use both hands) shall not be admitted
- Requests for transfer from other departments/disciplines shall be handled based on the relevant university regulation/senate legislation

6.2. Special Admission

Special admissions may be granted to attract potentially resourceful candidates such as mature students. Such special admissions shall be decided based on relevant university specific legislations and guidelines in line with the National direction.

7. Duration of the Study

- The bachelor degree in pharmacy requires 5 years or 10 semesters out of which the last 2 semesters (Year V) are dedicated to clerkships and community-based team training program.
- The curriculum contains 32 modules with a total of **355 ECTS**. The total ECTS is excluding two non-credited courses.

8. Graduation Requirements

Graduation requirement will be according to the rules and regulations of each university. Moreover, the following requirements must be fulfilled for a student enrolled in the B.Pharm program to be eligible for graduation:

- Has taken all the required modules for the program
- Obtained a minimum cGPA of 2.00
- Has not scored 'F' grade in any course,
- Has not scored less than 'C' grade in any of the major professional courses (course code starting with 'Phar...')
- Has not scored 'D' or lesser on any of the supportive courses/modules
- Has successfully completed and passed clerkship/professional practice program
- Has carried out a student research project on a selected and agreed topic of research problem and scored a minimum of 'C' grade
- Has passed a written comprehensive exit examination to be set for the program. After completion of all deductive courses, research and clerkships in good academic standing, students must take the comprehensive exit exam and score 50% or higher to be eligible for graduation. The exit exam shall be graded as Pass/Fail and a pass mark is considered as a prerequisite to sit for the national licensure exam.
- A student who fails to score a passing mark (50%) in the first comprehensive exit exam will be allowed to re-sit for the next exam after 3 months preparation with an appropriate academic support. The detailed conditions for remediation shall be in line with the National Remediation Guidelines to be introduced by the Ministry of Science and Higher Education and the Ministry of Health.

Attendance: Students should attend 100% of both lecture classes and practical attachments. The case of students who fail such requirements will be entertained as per senate legislation of the university and preceptor manual of the respective department.

9. Degree Nomenclature

- Up on successful completion of this program the graduate will be awarded “The Degree of Bachelor of Pharmacy (B.Pharm)” in English and “የ ባ ች ለ ር ዲ ግ ሪ ባ ፋ ር ማሲ ” in Amharic.

10. Teaching Methodology Mode of delivery

The program uses a Parallel mode of delivery.

Method of teaching

The following methods will be used as strategies to teach the courses within the different modules in the program:

- Lectures
- Laboratory practices
- Demonstrations
- Group works
- Home study
- Seminars
- Tutorials
- Home take assignments
- Skill lab
- Hospital attachment
- Health center attachment
- Pharmaceutical industries attachments
- Community based learning
- Field supervision
- Team training program
- Research project
- Role play
- Symposium
- Talk show
- Drama etc.

11. Assessment

Evaluation will be in the form of formative and summative assessment

- Formative assessment consists of,
 - 11.%.1. Continuous assessment (test, quiz, case presentation, assignment /individual – group exercise)
 - 11.%.2. Bed side, round
 - 11.%.3. Supervision
 - 11.%.4. Log book
 - 11.%.5. Portfolio
 - 11.%.6. Assessment of practical skills
- Summative assessment consists of:
 - 11.%.1. Final written exam
 - 11.%.2. Objective Structured Clinical Examination (OSCE)
 - 11.%.3. Oral exam
 - 11.%.4. Exit exam

N.B:

- Grading for exit exam will be P/F, and the student should score greater than or equal to 50% to pass.

12. Grading System

The program uses criteria referenced method of evaluation to value the performance of students at the end of each semester and/or year. The instructor uses the scale fixed and grade based on the pre-settled criteria, which are driven from the learning objectives of the topic/course. Thus, we do not use norm-referenced method.

Accordingly, students will be graded on the letter grade as well as on percentage grading system. The grading subsystem has to be adjusted to calculate the GPA based on the fixed scale range shown in the following table. The grading system that has been agreed upon by all public Schools of Pharmacy and subsequently approved by the then MOSHE shall be observed. However, the following table provides guidance on the raw marks and their corresponding fixed number and letter grades.

The grading of the comprehensive exit exam and the physical fitness courses will be on a “Pass” or “Fail” basis.

Raw Mark Interval [100%]	Corresponding fixed number Grade	Corresponding Letter Grade
[90, 100]	4.00	A+
[85, 90)	4.00	A
[80, 85)	3.75	A-
[75, 80)	3.50	B+
[70, 75)	3.00	B
[65, 70)	2.75	B-
[60, 65)	2.50	C+
[50, 60)	2.00	C
[45, 50)	1.75	C-
[40, 45)	1.00	D
< 40	0.00	F

Promotion criteria

- A student who scored “F” grade on a pre-requisite course shall not be allowed to take the next course.
- A student who scored “C-” and/or “D” in one or two major professional courses can be allowed to take remedial exam once. The student shall repeat the course/s if he/she could not score “C” or above on the remedial exam.
- A student who scored “C-” or below in three or more major professional courses shall not be promoted to the next semester, irrespective of his/her cGPA. However, a student who scored “C-” in only three major professional courses can be allowed to take remedial exam

for one of the courses and can be promoted if he/she scored a minimum grade of “C” in the remedial exam.

- Irrespective of his/her cGPA, a student is expected to score a minimum of “C” grade on all major professional courses to be eligible for the final year clerkship attachment.

13. List of Modules/Courses

13.1. Module and Course Code Assignment

In the module coding:

- The alphabets indicate to which program/department the module belongs
- The numbers should be four digits and
- The first number indicates year of study (starting from 1- for 1st year, 2- for 2nd year, etc.)
- The 2nd and 3rd numbers indicate module number (a two digit code starting from 01, 02...etc.)
- The last number indicates module category (core=1, elective=2, basic=3, general=4)

In the course coding:

- The alphabets indicate the course hosting department/program
- The numbers should be four digits
- The first number indicates year of study (starting from 1- for 1st year, 2- for 2nd year, etc.)
- The 2nd and 3rd numbers indicate module number (a two digit code starting from 01, 02...etc.)
- The last number indicates the course number (starting from 1, 2...etc.) within the module

13.2. List of Modules/Courses

During the allocation of credit points and sequencing of modules/courses assumption was made that all semesters other than Year I, Semester I, shall have duration of 20 weeks for the effective implementation of the curriculum.

The following tables show the list of modules and courses.

Major Compulsory (227 ECTS).

Module Number	Module Category	Module title	Module code	Course Code	Course Title	Course ECTS	Pre-requisite/ Co-requisite
7	Core	Introductory Pharmacy Module	Phar-M2071	Phar 2071	Introduction to pharmacy	2	None
				Phar2072	Pharmaceutical calculations	2	None
8	Core	Pharmacognosy and Alternative Medicine I	Phar-M2081	Phar2081	Chemistry of Natural products	5	Organic Chemistry, Organic Chemistry Laboratory
				Phar2082	Pharmacognosy	7	Chemistry of Natural Products
9	Core	Dosage form Sciences I	Phar-M2091	Phar2091	Integrated physical pharmacy and pharmaceutics-I	7	Pharmaceutical Calculations
			Phar-M2091	Phar2092	Practical Integrated physical pharmacy and pharmaceutics-I	2	Pharmaceutical Calculations & Integrated Physical Pharmacy and Pharmaceutics I
10	Core	Pharmacology module I	Phar-M2101	Phar2101	Pharmacology-I	7	Biochemistry I and II Physiology I and II Human Anatomy
11	Core	Medicinal Chemistry module I	Phar-M2111	Phar2111	Medicinal chemistry-I	7	Organic Chemistry/ Pharmacology I
12	Core	Dosage form Sciences II	Phar-M3121	Phar3121	Integrated physical pharmacy and pharmaceutics-II	7	Integrated Physical Pharmacy and Pharmaceutics I
				Phar3122	Practical Integrated physical pharmacy and pharmaceutics-II	2	Integrated Physical Pharmacy and Pharmaceutics II
13	Core	Pharmacology module II	Phar-M3131	Phar3131	Pharmacology II	7	Pharmacology I
				Phar 3132	Clinical toxicology	3	Pharmacology I and II
14	Core	Medicinal Chemistry module II	Phar-M3141	Phar3141	Medicinal chemistry-II	5	Medicinal Chemistry I Pharmacology II
15	Core	Pharmaceutical Analysis	Phar-M3151	Phar3151	Pharmaceutical analysis-I	7	None
				Phar3152	Pharmaceutical analysis-II	7	Pharmaceutical Analysis I
16	Core	Pharmacotherapeutics Module I	Phar-M3161	Phar3162	Integrated therapeutics-I	7	Pharmacology I
				Intm 3161	Physical assessment	2	Integrated Therapeutics I
				Phar3163	Integrated therapeutics-II	7	Integrated Therapeutics I
17	Core	Pharmaceutical Technology I	Phar-M3171	Phar3171	Immunological and biological products	3	Integrated Physical Pharmacy and Pharmaceutics I and II

18	Core	Social and administrative pharmacy module I	Phar-M3181	Comh3181	Health service management and policies	5	None
19	Core	Biopharmaceutics and Clinical Pharmacokinetics	Phar-M3191	Phar3191	Biopharmaceutics and Clinical Pharmacokinetics	7	Physiology II and Pharmacology II

Module Number	Module Category	Module title	Module code	Course Code	Course Title	Course ECTS	Pre-requisite/ Co-requisite
20	Core	Pharmaceutical Technology II	Phar-M4201	Phar4201	Industrial pharmacy	7	Integrated Physical Pharmacy and Pharmaceutics I and II
21	Core	Social and administrative pharmacy module II	Phar-M4211	Phar4211	Introduction to Pharmacoeconomics	3	None
				Phar4212	Pharmaceutical Supply Chain management	7	Health Service Management and Policies (PUBH3181)
				Phar4213	Medical supplies, equipment and reagents	3	None
				Phar4214	Pharmaceutical Marketing and promotion	3	None
22	Core	Pharmacotherapeutics Module II	Phar-M4221	Phar4221	Integrated therapeutics-III	7	Integrated Therapeutics I
				Phar4222	Integrated therapeutics-IV	7	Integrated Therapeutics I
23	Core	Pharmacognosy and Alternative Medicine II	Phar-M4231	Phar4231	Complementary and alternative medicine	3	Pharmacognosy
24	Core	Pharmacy practice module	Phar-M4241	Phar4241	Drug informatics	3	None
				Phar4242	Communication skills for pharmacists	3	None
				Phar4243	Pharmacy law and ethics	3	None
				Phar4244	Pharmacy practice	7	None
				Nurs4245	First aid	3	Course on Anatomy and Physiology
				PUBH4246	Nutrition	3	None
26	Core	Pharmaceutical Research I	Phar-M4261	Phar4261	Research Methods	3	Epidemiology and biostatistics courses
27	Core	Pharmacy clerkship I	Phar-M5271	Phar5271	Ambulatory care clerkship	5	Successful completion of all course work
				Phar5272	Drug information service clerkship	3	Successful completion of all course work
				Phar5273	Internal medicine clerkship	7	Successful completion of all course work
				Phar5274	Hospital pharmacy clerkship	7	Successful completion of all course work
28	Core	Pharmacy clerkship II	Phar-M5281	Phar5281	Pediatric clerkship	7	Successful completion of all course work
				Phar5282	Gynecology, obstetrics and family planning clerkship	3	Successful completion of all course work

				Phar5283	Pharmaceutical Manufacturing clerkship	5	Successful completion of all course work
				Phar5284	Community pharmacy clerkship	5	Successful completion of all course work
30	Core	Pharmaceutical Research II	Phar-M5301	Phar5301	Directed study	5	Research methods
31	Core	Team Training Program	PUBH M5311	ComH5311	Team training program	7	Successful completion of all course work
32	Core	Comprehensive Exit Exam	Phar-M5321	Phar5321	Comprehensive exit exam	0	Successful completion of all course work and clerkship

Major Elective (10 ECTS)

Module Number	Module Category	Module title	Module code	Course Code	Course Title	Course ECTS	Pre-requisite/ Co-requisite
25	Elective	Professional elective module	Phar-M4252	Phar4251	Introduction to Pharmacoepidemiology	5	None
				Phar4252	Phytochemistry	5	-
				Phar4253	Pharmaceutical Manufacturing	5	Pharmaceutical Technology Module
				Phar4254	Pharmacogenetics	5	Pharmacology I and II
				Phar4255	Pharmaceutical Quality control and quality assurance	5	Pharmaceutical Analysis I & II
				Phar4256	Drug design and synthesis	5	Organic Chemistry, Medicinal Chemistry I and II
				Phar4257	Warehouse management	5	Pharmaceutical supply chain management
				Phar4258	Research in pharmacology	5	Pharmacology I and II, Clinical toxicology
29	Elective	Professional elective clerkship	Phar-M5292	Phar5291	Psychiatry clerkship	5	Successful completion of all course work
				Phar5292	Surgery clerkship	5	Successful completion of all course work
				Phar5293	Oncology & Hematology clerkship	5	Successful completion of all course work
				Phar5294	Ophthalmology & ENT clerkship	5	Successful completion of all course work
				Phar5295	Emergency Medicine Clerkship	5	Successful completion of all course work
				Phar5296	Dermatology clerkship	5	Successful completion of all course work

				Phar5297	Pharmaceutical quality control	5	Successful completion of all course work
				Phar5298	Pharmaceutical regulatory affairs	5	Successful completion of all course work
				Phar5299	Pharmaceutical wholesale & promotion	5	Successful completion of all course work

Supportive (52ECTS)

Module Number	Module Category	Module title	Module code	Course Code	Course Title	Course ECTS	Pre-requisite/ Co-requisite
3	Basic	Biomedical sciences-I	Biom-M1033	Anat1031	Human Anatomy and Histology	7	None
				Phyl1032	Human Physiology-I	5	None
4	Basic	Chemistry	Chem-M1043	Chem1041	Organic chemistry	5	None
				Chem1042	Organic chemistry laboratory	2	None
5	Basic	Biomedical sciences-II	Biom-M2053	Path2054	Pathology	5	Physiology I&II , Biochemistry I&II
				Phyl2055	Human Physiology-II	5	Human Physiology I
				Bioc2051	Biochemistry-I	5	Organic Chemistry
				Bioc2052	Biochemistry-II	5	Organic Chemistry
				Mbio2053	Microbiology, Immunology and Parasitology	7	Physiology I&II , Biochemistry I&II Human Anatomy and Histology
6	Basic	Biostatistics and Epidemiology	PUBH2063	ComH2061	Biostatistics	3	None
				ComH2062	Epidemiology	3	None

General Education (66 ECTS)

Module Number	Module Category	Module title	Module code	Course Code	Course Title	Course ECTS	Pre-requisite/ Co-requisite
1	General	English Language Skills	Enla-M1014	FLEn1011	Communicative English Skill I	5	None
				FLEn1012	Communicative English Skill II	5	None
1	General	Social sciences and humanities		Psch 1011	General Psychology	5	None
				LoCT 1011	Critical Thinking	5	None
				GeES1011	Geography of Ethiopia & the Horn	5	None
				Anth 1012	Social Anthropology	4	None
1	General			Math 1011	Math for Natural Sciences	5	None
1	General			Phyc-1011	General Physics	5	None
1	General			SpSc 1011	Physical Fitness	0	None
1	General			MCiE 1012	Moral and Civic Education	4	None
1	General			SNIE 1012	Inclusiveness	4	None
1	General			MGMT 1012	Entrepreneurship	5	None
1	General			GlTr 1012	Global Trends	4	None
2	General			EmTe 1012	Introduction to Emerging Technologies	5	None
	General			Econ 1011	Economics	5	None

13.3. Module Sequencing by Semester and Year Year I Semester I

Module code	Module title	Course Code	Course Title	Course ECTS	Mode of delivery	Duration (hrs)	Lecture+ Tutorial+ Lab per week (hrs)
		FLEn1011	Communicative English Skill I	5	Parallel	135	3+0+0
		Psyc 1011	General Psychology	5	Parallel	135	2+1+0
		LoCT1011	Critical Thinking	5	Parallel	135	3+0+0
		GeES1011	Geography of Ethiopia & the Horn	5	Parallel	135	3+0+0
		Math1011	Math for Natural Sciences	5	Parallel	135	2+1+0
		Phyc 1011	General Physics	5	Parallel	135	2+1+0
		SpSc 1011	Physical Fitness	NC*=2ECTS	Parallel	81	*2 contact hours
Semester Total				30 ECTS*		891	

*** There will be a total ECTS of 33 if the NC Physical Fitness course is added**

Year I Semester II*

Module code	Module title	Course Code	Course Title	Course ECTS	Mode of delivery	Duration	Lecture+ Tutorial+ Lab per week (hrs)
		FLEn1012	Communicative English Skill II	5	Parallel	135	3+0+0
		EmTe 1012	Introduction to Emerging Technologies	5	Parallel	135	12+4+0
		Anth 1012	Social Anthropology	4	Parallel	108	3+0+0
		MCiE 1012	Moral and Civic Education	4	Parallel	108	4+0+0
Biom-M1033	Biomedical sciences-I	Anat1031	Human Anatomy and Histology	7	Parallel	189	4+1+2
Biom-M1033	Biomedical sciences-I	Phyl1032	Human Physiology-I	5	Parallel	135	3+1+0
Chem-M1043	Chemistry	Chem1041	Organic chemistry	5	Parallel	135	12+4+0
Chem-M1043	Chemistry	Chem1042	Organic chemistry laboratory	2	Parallel	54	0+0+9
Semester Total				37 ECTS		999	

*** 20 Weeks semester**

Year II Semester I*

Module code	Module title	Course Code	Course Title	Course ECTS	Mode of delivery	Duration	Lecture+ Tutorial+ Lab per week (hrs)
Biom-M2053	Biomedical sciences-II	Path2054	Pathology	5	Parallel	135	5+3+0
Phar-M2081	Pharmacognosy and Alternative Medicine I	Phar2081	Chemistry of Natural products	5	Parallel	135	3+0+0
Biom-M2053	Biomedical sciences- II	Phyl2055	Human Physiology-II	5	Parallel	135	2.5+1+0
Biom-M2053	Biomedical sciences- II	Bioc2051	Biochemistry-I	5	Parallel	135	5+3+0
Biom-M2053	Biomedical sciences- II	Bioc2052	Biochemistry-II	5	Parallel	135	5+3+0
Biom-M2053	Biomedical sciences-II	Mbio2053	Microbiology, Immunology and Parasitology	7	Parallel	189	8+3+6
Phar-M2071	Introductory Pharmacy Module	Phar 2071	Introduction to pharmacy	2	Parallel	54	1+0+2
Phar-M2071	Introductory Pharmacy Module	Phar2072	Pharmaceutical calculations	2	Parallel	54	1+3+0
		Econ 1011	Economics	5	Parallel	135	3+0+0
Semester Total				41 ECTS		1107	

* 20 Weeks semester

The following were the considerations for amendment:

- Universities, which have **not** offered the Organic Chemistry Lab course in the second semester of the first year, should arrange to offer the course in the first week of Year II-Semester I and retrospectively register the students for Year I-Semester II.
- Universities, which already completed McIe-1012, can replace it with a similar module of 3 – 5 ECTS.
- In case of universities, which already offered any one of the listed courses, it is important that they replace the course/s with an equivalent course, in order to avoid credit spillover to the later years/semesters.

Year II Semester II*

Module code	Module title	Course Code	Course Title	Course ECTS	Mode of delivery	Duration	Lecture+ Tutorial+ Lab per week (hrs)
Phar-M2081	Pharmacognosy and Alternative Medicine I	Phar2082	Pharmacognosy	7	Parallel	189	4 +1+3
Phar-M2101	Pharmacology module I	Phar2101	Pharmacology-I	7	Parallel	189	4+1+3
Phar-M2111	Medicinal Chemistry module I	Phar2111	Medicinal chemistry-I	7	Parallel	189	4+1+3
Phar-M2091	Dosage form Sciences I	Phar2091	Integrated physical pharmacy and pharmaceuticals-I	7	Parallel	189	4+1+0
Phar-M2091	Dosage form Sciences I	Phar2092	Practical Integrated physical pharmacy and pharmaceuticals-I	2	Parallel	54	0+0+3
PUBH2063	Biostatistics and Epidemiology	ComH2061	Biostatistics**	3	Parallel	81	2+0+0
PUBH2063	Biostatistics and Epidemiology	ComH2062	Epidemiology**	3	Parallel	81	2+0+0
		SNIE 1012	Inclusiveness	4	Parallel	81	2+0+0
Semester Total				40 ECTS		1053	

*** 20 Weeks semester**

**** In the event that the contents of the course History of Ethiopia & the Horn (Hist.1012) are completed before students start this semester, the course can be offered in place of Biostatistics and Epidemiology. Then Epidemiology can be offered in Year III Semester I or Year III Semester II and Biostatistics can be offered in Year III Semester II or Year IV Semester I.**

Year III Semester I*

Module code	Module title	Course Code	Course Title	Course ECTS	Mode of delivery	Duration	Lecture+ Tutorial+ Lab per week (hrs)
Phar-M3121	Dosage form Sciences II	Phar3121	Integrated physical pharmacy and pharmaceutics-II	7	Parallel	189	4+1+0
Phar-M3121	Dosage form Sciences II	Phar3122	Practical Integrated physical pharmacy and pharmaceutics-II	2	Parallel	54	0+0+3
Phar-M3131	Pharmacology module II	Phar3131	Pharmacology II	7	Parallel	189	4+1+3
Phar-M3141	Medicinal Chemistry module II	Phar3141	Medicinal chemistry-II	5	Parallel	135	3+1+0
Phar-M3151	Pharmaceutical Analysis	Phar3151	Pharmaceutical analysis-I	7	Parallel	189	4+1+3
Phar-M3161	Pharmacotherapeutics Module I	Phar3162	Integrated therapeutics-I	7	Parallel	189	4+1+1
Phar-M3161	Pharmacotherapeutics Module I	Intm 3161	Physical assessment	2**	Parallel	54	1+0+3
Semester total***				37 ECTS		999	

*** 20 Weeks semester**

**** The physical assessment course is co-requisite to Integrated therapeutics-I**

Year III Semester II *

Module code	Module title	Course Code	Course Title	Course ECTS	Mode of delivery	Duration	Lecture+ Tutorial+ Lab per week (hrs)
Phar-M3191	Biopharmaceutics and Clinical Pharmacokinetics	Phar3191	Biopharmaceutics and Clinical Pharmacokinetics	7	Parallel	189	4+1+0
Phar-M3171	Pharmaceutical Technology I	Phar3171	Immunological and biological products	3	Parallel	81	4 +2+0
Phar-M3131	Pharmacology module II	Phar 3132	Clinical toxicology	3	Parallel	81	2 +2+0
Phar-M3181	Social and administrative pharmacy module I	Comh3181	Health service management and policies	5	Parallel	135	4 +0+0
Phar-M3161	Pharmacotherapeutics Module I	Phar3163	Integrated therapeutics-II	7	Parallel	189	4+1+2
		MGMT 1012	Entrepreneurship	5	Parallel	135	3 + 1+ 0
Phar-M3151	Pharmaceutical Analysis	Phar3152	Pharmaceutical analysis-II	7	Parallel	189	4+1+3
Semester total**				37		999	

*** 20 Weeks semester**

Year IV Semester I*

Module code	Module title	Course Code	Course Title	Course ECTS	Mode of delivery	Duration	Lecture+ Tutorial+ Lab per week (hrs)
Phar-M4201	Pharmaceutical Technology II	Phar4201	Industrial pharmacy	7	Parallel	189	4+0.5+3
Phar-M4211	Social and administrative pharmacy module II	Phar4211	Introduction to Pharmacoeconomics	3	Parallel	81	2+2+0
Phar-M4211	Social and administrative pharmacy module II	Phar4212	Pharmaceutical Supply Chain management	7	Parallel	189	4+1+0
Phar-M4221	Pharmacotherapeutics Module II	Phar4221	Integrated therapeutics-III	7	Parallel	189	4+1+2
Phar-M4231	Pharmacognosy and Alternative Medicine II	Phar4231	Complementary and alternative medicine**	3	Parallel	81	2+1+1**
Phar-M4241	Pharmacy practice module	Phar4241	Drug informatics	3	Parallel	81	4+2+2
Phar-M4241	Pharmacy practice module	Phar4242	Communication skills for pharmacists	3	Parallel	81	4+2+2
Phar-M4241	Pharmacy practice module	Phar4243	Pharmacy law and ethics	3	Parallel	81	4 +1+0
Semester total				36 ECTS		972	

*** 20 Weeks semester**

**** Consider visits to botanical gardens/registered traditional healers or their settings/Health bureau regulating the traditional healers**

Year IV Semester II *

Module code	Module title	Course Code	Course Title	Course ECTS	Mode of delivery	Duration	Lecture+ Tutorial+ Lab per week (hrs)
		GITr 1012	Global Trends	4	Parallel	108	4+0+0
Phar-M4211	Social and administrative pharmacy module II	Phar4213	Medical supplies, equipment and reagents	3	Parallel	81	4+1+0
Phar-M4211	Social and administrative pharmacy module II	Phar4214	Pharmaceutical Marketing and promotion	3	Parallel	81	4+2+2
Phar-M4221	Pharmacotherapeutics Module II	Phar4222	Integrated therapeutics-IV	7	Parallel	189	4+1+2
Phar-M4241	Pharmacy practice module	Phar4244	Pharmacy practice***	7	Parallel	189	4 +0+6
Phar-M4241	Pharmacy practice module	Nurs4245	First aid	3	Parallel	81	4 +2+10
Phar-M4241	Pharmacy practice module	PUBH4246	Nutrition	3	Parallel	81	4 +2+0
Phar-M4252	Professional elective module	*	Professional elective course	5	Parallel	135	4+1+1
Phar-M4261	Pharmaceutical Research I	Phar4261	Research Methods	3	Parallel	81	4+1+0
Semester total****				38		1026	

*** 20 Weeks semester**

*****The course shall be owned by all departments/units in the school/department of pharmacy and the theory and practice should give enough time for learning and practicing on the various areas of pharmacy practice such as pharmaceutical manufacturing, regulatory affairs, quality control, supply chain management, etc.**

Year V Semester I and Year V Semester II

Final year attachment clerkship will be year-based. Student assessment will be made at the end of the year

Module code	Module title	Course Code	Course Title	Course ECTS*	Mode of delivery	Duration	Number of weeks
Phar-M5271	Pharmacy clerkship I	Phar5271	Ambulatory care clerkship	5	Parallel	135	3
Phar-M5271	Pharmacy clerkship I	Phar5272	Drug information service clerkship	3	Parallel	81	2
Phar-M5271	Pharmacy clerkship I	Phar5273	Internal medicine clerkship	7	Parallel	189	4
Phar-M5271	Pharmacy clerkship I	Phar5274	Hospital pharmacy clerkship	7	Parallel	189	4
Phar-M5281	Pharmacy clerkship II	Phar5281	Pediatric clerkship	7	Parallel	189	4
Phar-M5281	Pharmacy clerkship II	Phar5282	Gynecology, obstetrics and family planning clerkship	3	Parallel	81	2
Phar-M5281	Pharmacy clerkship II	Phar5283	Pharmaceutical Manufacturing clerkship	5	Parallel	135	3
Phar-M5281	Pharmacy clerkship II	Phar5284	Community pharmacy clerkship	5	Parallel	135	3
Phar-M5292	Professional elective clerkship	***	Elective attachment	5	Parallel	135	3
Phar-M5301	Pharmaceutical Research II	Phar5301	Directed study	5	Parallel	135	NA
PUBHM5311	Team Training Program	ComH5311	Team training program	7	Parallel	189	6
Phar-M5321	Comprehensive exit exam	Phar5321	Comprehensive exit exam	Non credit	Parallel	NA	3 weeks for preparation
Semester total				59 ECTS		1593	37**

* To be determined based on a student's preference

** A one-week off after major pharmacy clerkships is recommended and students are encouraged to plan their time so that they can **find extra time for data collection and/or experiments to fulfil the directed study requirements.**



14. Quality Assurance

The school/department quality assurance committee is responsible for the management and monitoring of the program. Both formative and summative evaluations will be done to assess the realization of the curriculum's objectives. The following mechanisms will be employed to evaluate whether or not the courses offered in the program meet the standards:

- A course syllabus according to the course content indicated in this curriculum should be prepared for each course with time frame.
- The respective School/departments will evaluate the agreement between examination contents and the course syllabi.
- Recruitment of qualified staff
- Examination and continuous assessment (formative and summative)
- Periodic acquisition of up – to – date references, laboratory equipment and reagents
- Supervised practices in the training health facilities
- Periodic evaluation of the curriculum and the program in general
- Periodic evaluation and revision of the curriculum based on the feedback from stakeholders/employers, graduates and students.

15. Resource Profile

- **Names and ranks of active teaching and technical staff**

